CLINICAL PRESENTATION & CASE ANALYSIS

Hemodialysis Care to a Patient With End-Stage Renal Disease (ESRD)

A comprehensive clinical evaluation of renal failure pathophysiology, dialyzer fluid dynamics, lab profiles, and evidence-based nursing interventions.

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Introduction & Definition of Hemodialysis

What is Hemodialysis (HD)?

Hemodialysis is an extracorporeal life-sustaining therapy that removes accumulated metabolic uremic wastes and excess fluid from the circulatory system when native renal function fails.

Extracorporeal Circuit: Blood is pumped via vascular access through sterile synthetic tubing to a hemodialyzer machine containing a semi-permeable membrane ("artificial kidney"). Once purified, blood safely returns to systemic circulation.

Clinical Delivery Settings & Regimen

- **In-Center Inpatient / Outpatient Unit:** Typical treatment duration is **3 to 5 hours** per session, performed **3 times per week**.
- **Home Hemodialysis:** Tailored nocturnal or short daily schedules under specialized patient training.
- **Extracorporeal Volume:** Only ~200–250 mL of blood resides outside the body at any single moment during steady-state filtration.

Dialyzer Mechanics & Core Physics



Dialysate fluid flows countercurrent to blood flow across hollow-fiber capillary membranes, maximizing the diffusion gradient across the semi-permeable barrier.

Case Study: Mrs. D.K. (76-Year-Old Female)

Admission Background & Chief Complaint

Patient: Mrs. D.K., a 76-year-old female admitted to the **Gen 4 Medicine** inpatient unit with an admission diagnosis of "*failure to cope at home*".

Past Medical History: End-Stage Renal Disease (ESRD) on maintenance hemodialysis, Type 2 Diabetes Mellitus (T2DM), and Chronic Essential Hypertension.

Psychosocial & Functional Assessment

- Lives alone with limited domestic and community support networks.
- Increasing difficulty managing complex daily medication regimens and renal diabetic dietary restrictions.
- Severe ADL decline driven by chronic fatigue, generalized muscular weakness, and frequent exhausting medical transport appointments.

Baseline Admission Vital Signs

VITAL SIGN	PATIENT VALUE	CLINICAL ASSESSMENT
Blood Pressure	162/82 mmHg	Stage 2 Hypertension; significant risk for end-organ vascular injury in ESRD
Heart Rate	64 bpm	Normal resting sinus rate; hemodynamically stable
Respiratory Rate	20 breaths/min	Mild tachypnea at rest; assess for fluid overload/pulmonary congestion
Temperature	36.7 °C	Afebrile; no acute systemic hyperthermic response
SpO2 Saturation	96% on room air	Adequate baseline arterial oxygenation without supplemental support

Clinical Diagnostic Summary: Mrs. D.K. is afebrile and hemodynamically stable but exhibits severe uncontrolled systolic hypertension superimposed on chronic uremia and fluid-handling compromise.

Comprehensive Admission Laboratory Profile

DIAGNOSTIC LAB TEST	PATIENT RESULT	STANDARD ADULT REFERENCE	CLINICAL NURSING INTERPRETATION
Hemoglobin (Hgb)	98 g/L	120 – 160 g/L (female)	**Low – Normocytic Anemia of CKD:** Secondary to renal erythropoietin (EPO) synthesis deficiency and reduced RBC survival.
Platelets (Plt)	82 ×10&sup9;/L	150 – 400 ×10&sup9;/L	**Low – Thrombocytopenia:** Uremic platelet dysfunction elevates post-puncture bleeding risks at AV fistula sites.
Serum Creatinine	281 µmol/L	50 – 100 µmol/L (female)	**High – Severe Renal Clearance Failure:** Massive loss of nephron glomerular filtration rate (GFR < 15 mL/min).
Blood Urea Nitrogen (Urea)	51.4 mmol/L	2.5 – 7.8 mmol/L	**Critical High – Overt Uremia:** Manifests clinically with severe nausea, pruritus, neurological lethargy, and anorexia.
Serum Albumin	27 g/L	35 – 50 g/L	**Low – Protein-Energy Wasting:** Chronic systemic inflammatory state, malnutrition, and impaired oncotic pressure.
Magnesium	1.16 mmol/L	0.7 – 1.0 mmol/L	**High – Hypermagnesemia:** Decreased renal excretion; risk for reduced deep tendon reflexes and cardiac conduction delay.
Serum Sodium (Na+)	132 mmol/L	135 – 145 mmol/L	**Low – Mild Dilutional Hyponatremia:** Result of impaired free-water clearance and extracellular volume expansion.

Priority Nursing Correlates

Uremic Syndrome: BUN of 51.4 mmol/L explains patient's severe exhaustion, confusion, and failure to thrive at home.

Electrolyte & Anemia Risk: Hgb 98 g/L requires ESA/iron coordination; calcium/magnesium imbalances mandate cardiac telemetry and seizure precautions.

Target Clearance: Immediate hemodialysis required to clear urea, correct metabolic acidosis, and optimize volume.

Principles of Dialysis Clearance: Diffusion & Osmosis

1. Diffusion (Solute Clearance)

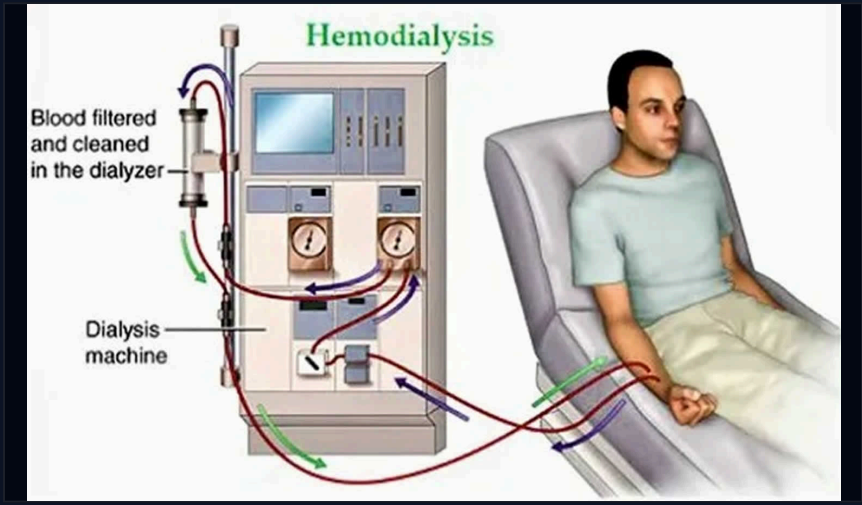
The spontaneous movement of small dissolved solutes (urea, creatinine, potassium) from an area of high concentration in the patient's blood across a semipermeable membrane into a lower concentration in dialysate bath.

- Dialysate contains physiological electrolytes (bicarbonate, low potassium) to restore acid-base and chemical equilibrium.

2. Osmosis (Fluid Shift)

The movement of solvent water molecules through a semi-permeable membrane from lower solute concentration to higher solute concentration, driven by osmotic gradients.

Semi-Permeable Membrane Dynamics



Synthetic hollow-fiber dialyzer capillaries selectively allow passage of small molecules (<500 Da) while blocking essential high-molecular-weight proteins (albumin) and blood cells (erythrocytes, leukocytes).

Why Dialysis is Needed: The "AEIOU" Framework

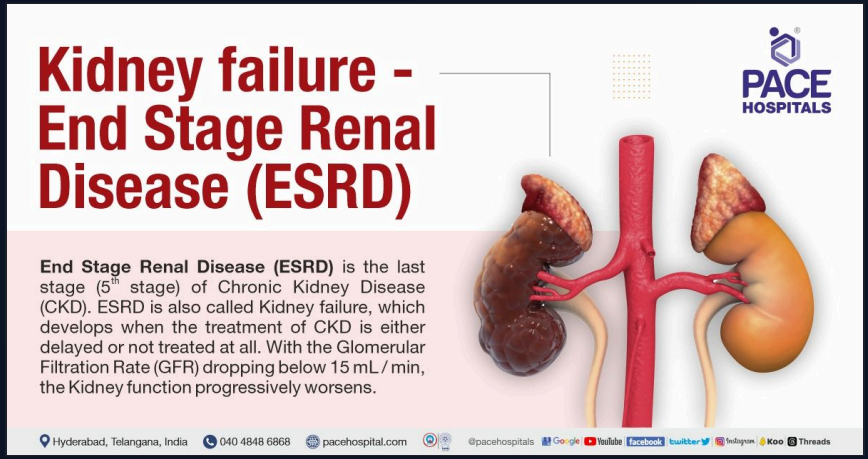
Clinical Rationale for Renal Replacement Therapy

Native kidneys normally execute crucial regulatory, excretory, and endocrine roles:

- Filtration of toxic end-products of nitrogenous metabolism (urea, uric acid, creatinine).
- Precise homeostasis of extracellular fluid volume and vascular pressures.
- Regulation of systemic serum electrolytes (K+, Na+, Ca2+, $PO4^{3}-$, Mg2+).
- Systemic acid-base buffering via hydrogen ion excretion and HCO3− reabsorption.

When End-Stage Renal Failure is reached (GFR < 15 mL/min/1.73m²), conservative medical management fails to prevent fatal toxic uremic accumulation and hyperkalemic cardiac arrest.

Emergency Nephrology Mnemonic: AEIOU



- **A – Acidosis:** Intractable metabolic acidosis (pH < 7.10) resistant to bicarbonate.
- **E – Electrolytes:** Severe refractory hyperkalemia (K+ > 6.5 mmol/L) with ECG changes.
- **I – Ingestion / Intoxication:** Toxic dialyzable substances (lithium, ASA, toxic alcohols).
- **O – Overload:** Acute pulmonary edema unresponsive to aggressive intravenous diuretics.
- **U – Uremia:** Symptomatic uremic encephalopathy, pericarditis, or

Learning Goal: Hemodialysis & CKD Management

Explicit Clinical Learning Objective

> **"I will master the activity and synthesize the core biophysical purpose of hemodialysis, analyze its mechanical function in the body, and articulate critical nursing safety implications for patients with advanced chronic kidney disease (CKD)."**

Competency Alignment (CNO Standard): Integrates physiological assessment, safe medication administration, vascular access preservation, and client-centered discharge coaching.

Core Performance Competencies Evaluated

- Assessment and bruit/thrill auscultation of arteriovenous (AV) fistulas / grafts.
- Calculation of interdialytic fluid weight gains and ultrafiltration targets.
- Safe management of dialyzable medications (timing anti-hypertensives).
- Patient education on fluid restrictions (<1.0 L/day) and low-potassium/phosphorus nutrition.

Digital Learning & Multimedia Simulation

Interactive procedural review of blood circuit, pressure transducers, heparinization protocols, and air embolus sensor alarms.

Extracorporeal Circuit Walkthrough

Focus: Safety mechanisms, bubble trap chambers, dialysate heating, and transmembrane pressure (TMP) monitors.

Continuous peer and preceptor demonstration ensures student nurse readiness in complex nephrology environments.

Pathophysiology & Systemic Nursing Insights

Multi-System Cascade of ESRD

End-Stage Renal Disease (ESRD) represents the terminal, irreversible stage of chronic kidney disease characterized by massive nephron sclerosis and destruction.

- **Glomerular Hyperfiltration & Fibrosis:** Chronic diabetic microvascular disease and endothelial shear stress destroy capillary loops.
- **Endocrine Failure:** Cessation of renal erythropoietin leads to severe non-regenerative anemia; failure to hydroxylate 25(OH)D to calcitriol triggers secondary hyperparathyroidism and bone demineralization (renal osteodystrophy).
- **Cardiovascular Overload:** Volume retention causes concentric left ventricular hypertrophy (LVH), congestive heart failure, and calciphylaxis.

Evidence-Based Nursing Implications (Supramanian et al., 2024)

> **Systemic Multi-Organ Surveillance:** Nurses must anticipate pulmonary crackles, peripheral third-spacing, cognitive blunting, uremic pruritus, and hyperkalemic dysrhythmias.

> **Vascular Access Protection:** Strict ban on venipunctures, blood pressures, and tight sleeves on the access extremity to prevent fistula thrombosis and pseudoaneurysm.

> **Medication Stewardship:** Rigorous dose adjustments for renally eliminated drugs (e.g., gabapentin, enoxaparin, vancomycin) to avoid severe iatrogenic toxicity.

Ultrafiltration (UF): Fluid & Waste Removal Dynamics

Ultrafiltration Principles & Transmembrane Pressure

Ultrafiltration is the convective movement of fluid across a semi-permeable membrane driven by an artificial hydrostatic pressure gradient (Transmembrane Pressure / TMP) created by dialysis machine effluent pumps.

- Unlike simple diffusion, UF physically pulls bulk fluid along with dissolved small and middle solutes ("solvent drag").
- **Dry Weight Concept:** The patient's ideal post-dialysis weight at which they are normotensive without peripheral edema or pulmonary congestion.

Interdialytic Weight Gain (IDWG) Limits

For safety, fluid removal during a single 4-hour session should rarely exceed **1.0 – 1.5 L/hr** (or ~10–13 mL/kg/hr) to prevent sudden hypovolemic intravascular collapse and myocardial stunning.

Intradialytic Complications & Nursing Actions

COMPLICATION	UNDERLYING MECHANISM	IMMEDIATE NURSING ACTION
Hypotension	Excessive UF rate exceeding vascular refilling capacity	Lower UF rate; place patient in Trendelenburg position; administer 100–200 mL 0.9% NaCl bolus as ordered.
Muscle Cramps	Rapid shifts in extracellular osmolarity and tissue perfusion	Bolus normal saline; gently stretch affected extremity; re-evaluate dry weight prescription.
Disequilibrium	Rapid BUN clearance creates osmotic cerebral edema	Slow blood flow rate (BFR); stop treatment if neurological deterioration occurs; notify nephrologist.

Comprehensive Nursing Plan of Care: Mrs. D.K.

Pre-Dialysis Assessment Checklist

- **Weight & Fluid Calculation:** Weigh patient on calibrated scale; compute exact difference from dry weight to establish target UF goal.
- **Vascular Access Integrity:** Palpate continuous thrill, auscultate vascular bruit; inspect for erythema, purulence, hematoma, or limb swelling.
- **Medication Timing:** Hold morning antihypertensives (e.g., ACEi, beta-blockers) to prevent profound intradialytic hypotension; hold dialyzable water-soluble vitamins.

Post-Dialysis Re-Assessment

- Check post-treatment weight and orthostatic vitals; ensure bleeding at cannulation sites has ceased before dressing application.

Patient & Family Discharge Education

Dietary Coaching: Limit sodium (<2 g/day), potassium (<2–3 g/day), and phosphorus (<800–1000 mg/day). Take phosphate binders with first bite of every meal.

Fluid Restrictions: Maintain strict 1000–1500 mL/day fluid intake; educate on using measured water bottles and ice chips to manage thirst.

Home Support Coordination: Refer to community care access networks (Home and Community Care Support Services) for transportation, medication blister packaging, and personal support workers.

References & Clinical Literature

Peer-Reviewed Citations & Clinical Guidelines

- **College of Nurses of Ontario (CNO). (2023).** *Code of Conduct & Professional Standards for Registered Nurses and Registered Practical Nurses*. Toronto, ON: CNO.
- **Medical Advisory Board. (2025, August 12).** *What is ultrafiltration in hemodialysis?* Definitions and physics of transmembrane convective clearance across synthetic dialyzer membranes.
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- **Ontario Laboratory Information System (OLIS). (2026).** *Standard Reference Ranges for Inpatient Clinical Chemistry and Hematology*. Ontario Ministry of Health.
- **Supramanian, K., Sekar, M., & Safwan Hadi Nor Afendi, N. (2024).** *Chronic Kidney Disease: Etiology, Pathophysiology, and Management Strategies to Increase Quality of Life*. In: Chronic Kidney Disease – Novel Insights into Pathophysiology and Treatment. IntechOpen. doi:10.5772/intechopen.1005083.